

Monics and epics

Another example of doing things categorically, that is, moving away from elements and expressing everything in terms of morphisms.

There is a general process of “doing” category theory like this:

Doing category theory, type 1

1. Become curious about a structure somewhere, often in sets and functions.
2. Express it categorically, that is, using the objects and arrows of the category **Set**, never referring to elements of sets.
3. Take the definition to other categories to see what it corresponds to there.

The idea is that as basic math happens in **Set**, it could be fruitful to take any feature that we often use in there and look for it in other categories, so that we can transfer our techniques from **Set** to somewhere else.

When studying category theory in its own right, without necessarily going through the usual preliminary stages of math first, sometimes this process gets turned around because you might not have seen the supposedly “motivating” phenomenon in **Set** ever before in your life. It’s like the classic research seminar where the speaker tells you that something you’ve never heard of before is just an example of this other thing you’ve never heard of before.[†]

If you haven’t met the “motivating” examples before, the categorical structure may instead have its own intrinsic motivation. In that case we’re bypassing step 1 in the above scheme, and doing something more like this:

Doing category theory, type 2

1. Fit some logical pieces together to create an interesting-looking structure in a general category.
2. Take the definition to some examples of categories to see what it gives.

I think of the first type of approach as “externally motivated” whereas the second type is “internally motivated” as we’re motivated entirely from within category theory by something that fits together in an abstractly interesting way,

[†] I believe I heard this succinct description of seminars from John Baez.

and we then see where we can use it, rather than starting with a specific use as a goal.

I have written before[†] about how this is like two different ways of being a tourist in a new city. The external way is to decide on the places you want to visit, and then work out how to get to them. The internal way is to just plonk yourself in the middle of the city and start following your nose. Of course one often does a mix of the two: perhaps you decide on one place to visit to start the day and then follow your nose from there. The “following your nose” approach works better in some cities than others.

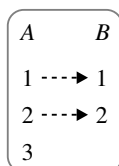
We are going to continue from the previous chapter where we were looking at isomorphisms of sets. When we characterized these as bijections the definition implicitly had two parts, giving two distinct ways in which a function might fail to be a bijection; this is in turn related to the asymmetry in the definition of a function. In this chapter we’re going to focus on those two aspects of functions, and see what abstract structure they translate to in category theory.

15.1 The asymmetry of functions

When we first introduced functions we drew some pictures showing inputs going to outputs and observed that some features aren’t allowed. Those features were not symmetrical. Here they are in pictures.

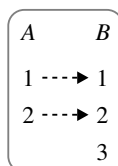
Every input must produce an output. So inputs with no arrow attached are not allowed.

not allowed



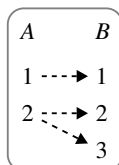
There can be outputs that are not achieved. So outputs with no arrow attached are allowed.

allowed



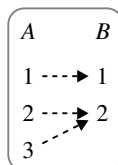
One input cannot produce multiple outputs. So arrows meeting on the left are not allowed.

not allowed



One output can be achieved by multiple inputs. So arrows meeting on the right are allowed.

allowed



While this asymmetry is fine for functions in general, it is what prevents general functions from being isomorphisms. This makes sense as isomorphism is a generalization of symmetry of relations.

[†] *How to Bake Pi*, Profile Books (Basic Books in the US), 2015.

An isomorphism of sets is when the inputs and outputs match perfectly, with arrows neither meeting up nor omitting objects. It gives pictures such as the one on the right.

A	B
1	1
2	2
3	3

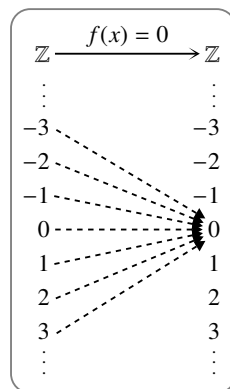
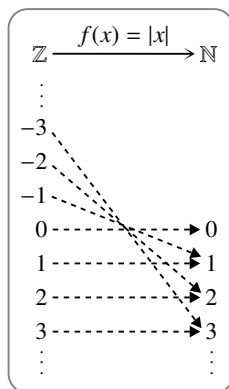
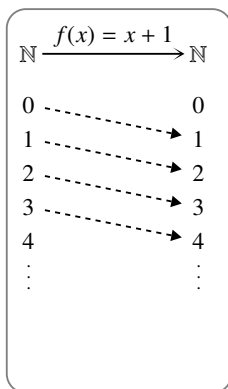
Mathematicians take some interest in the features that *prevent* this from happening, and look more closely at how to iron them out.

Things To Think About

T 15.1 Think about the following functions, which are not isomorphisms. Exactly which of the two pertinent features is getting in the way (or is it both)? Note that although these sets are infinite you could still draw a picture showing the general features, which might help you.

1. $f: \mathbb{N} \rightarrow \mathbb{N}$ where $f(x) = x + 1$.
2. $f: \mathbb{Z} \rightarrow \mathbb{N}$ where $f(x) = |x|$. This function takes the *absolute value* of x , that is, it ignores any negative signs.
3. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ where $f(x) = 0$.

Here are pictures of the general pattern of each function.



For the first function we see that the output 0 is not hit by any input, so the function is not invertible. The function “+1” is invertible on the integers, but the problem here is that we have no negative numbers. In fact the problem is really that we have a *smallest* number, and this output will never be hit by the “+1” function as long as you have the same set of numbers on both sides.

The second function fails to be invertible because we have multiple inputs going to the same output, that is, arrows meet up on the right. The third function is quite extreme and exhibits both of these problems: there are outputs with no

arrow landing on them, and also multiple arrows meeting at the same output. In fact *all* the arrows meet at the same output. This is what happens when we multiply by 0, and shows why we can't invert that process or "divide" by 0.

We will now look at each of those problems separately.

15.2 Injective and surjective functions

Isomorphisms are the best-behaved type of function, where both types of problematic behavior are ruled out. But there are "in between" type functions where just one type has been ruled out. If you already know about injective and surjective functions you could go through this section quickly, just to pick up the spirit of how I like to think of these things, before moving on to the categorical approach.

Injective functions

A function is called *injective* (or an injection) if it does not have the problem of "arrows meeting on the right". There are various equivalent ways of thinking about this and it's good to be able to think via any of them.

- Arrows do not meet on the right.
- Every output is hit at most once.
- If an output is achieved, there is a unique input that went to it.
- Given any output there is *at most* one input that goes to it.
- If two inputs go to the same output they must have been the same already.[†]

The last way of thinking takes us to the formal definition.

Definition 15.1 A function $f : A \rightarrow B$ is called *injective* if

$$\forall x, y \in A, f(x) = f(y) \implies x = y.$$

Injective functions are sometimes called "one-to-one", perhaps on the grounds that it is a more intuitive term to try and remember. However I don't like that terminology because I think (to my intuition, anyway) that makes it sound like a perfect correspondence, that is, a bijection.

The formal definition is arguably not intuitive at all, but is in a convenient form for doing rigorous proofs. It comes back to that principle of proving something is unique by assuming there are two and showing they are the same.

[†] Remember that "two inputs" in math can be referring to the same input twice; we didn't say "two distinct inputs".

For example, if we take $f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(x) = x + 1$, here's how we could prove that the function is injective:

$$f(x) = f(y) \text{ means } x + 1 = y + 1, \text{ which implies } x = y.$$

To prove that a function is *not* injective we have to prove the negation[†] of the definition of injective, that is: $\boxed{\exists x, y \in A \text{ such that } f(x) = f(y) \text{ but } x \neq y.}$

For example if we define $f: \mathbb{Z} \rightarrow \mathbb{N}$ by $f(x) = |x|$, then we can show this is not injective by observing that $f(-1) = f(1)$ although $-1 \neq 1$. Translating between intuition and formality in math is one of the challenges, but if you can get your head around it this opens up huge worlds of complex and nuanced logical arguments that are too complicated to follow by intuition alone.

Surjective functions

A function is called *surjective* (or a surjection) if it does not have the problem of “outputs with no arrow to them”. Here are some equivalent ways of thinking about it.

- There are no outputs without an arrow to them.
- Every output is hit at least once.
- Given any output there is at least one input that goes to it.

The last one takes us to the formal definition.

Definition 15.2 A function $f: A \rightarrow B$ is called *surjective* if

$$\forall b \in B \quad \exists a \in A \text{ such that } f(a) = b.$$

Note on terminology

Surjective functions are sometimes called “onto” to encapsulate the idea that they take the inputs onto everything, but I always think it sounds strange to use a preposition as an adjective. The phrase “this function is onto” makes me want to go “onto what?”. I think this other terminology is there because of the idea that “injective” and “surjective” sound scary, or perhaps it’s difficult to remember which way round they are. I think of medical injections, and you definitely don’t have two needles going into the same spot. For “surjection” I think of the root “sur” as in “on”.

To prove that something is surjective we can consider an arbitrary element of B and then exhibit an element of A that lands on it. For example, if we consider $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x + 1$, we can show it is surjective like this:

$$\text{Given any } n \in \mathbb{Z}, \text{ we have } f(n - 1) = n.$$

[†] See Appendix B if you need help with negating statements like this.

That is, we took $b = n$ and $a = n - 1$ in the general form of the definition.

To show that something is not surjective we need to negate the definition, which gives us this: $(\exists b \in B \text{ such that } \forall a \in A \ f(a) \neq b.)$

For example if we take $f: \mathbb{N} \rightarrow \mathbb{N}$ and $f(x) = x + 1$ we can show this is not surjective as follows:

Given any $n \in \mathbb{N}$, we know $f(n) > 0$. Therefore $\forall n \in \mathbb{N}, f(n) \neq 0$.

Bijjective functions

Finally note that if a function is both injective and surjective then it is called *bijjective* (or a bijection). Here are some ways of bringing together the separate ideas of injective and surjective functions to make bijjective functions.

- Arrows neither *meet* on the right nor *leave anything out* on the right.
- Every output is hit *exactly* once.
- Given any output there is *exactly one* input that goes to it.

Combining the formal definitions we get this.

Definition 15.3 A function $f: A \rightarrow B$ is called *bijjective* if it is injective and surjective, that is: $(\forall b \in B \ \exists! a \in A \text{ such that } f(a) = b.)$

Here the exclamation mark $!$ is mathematical notation for “unique”.[†] Note that the uniqueness applies to everything after it: it’s not a unique element in A , it’s a unique “element in A such that $f(a) = b$ ”.

Here again we have “exactly one” broken down into “at least one” and “at most one”, or existence and uniqueness, and encapsulated by the $\exists!$ notation.

Examples

Injectivity and surjectivity are logically independent, that is, in general, functions can be injective or surjective or both or neither.[‡]

Things To Think About

T 15.2 Try and interpret the following situations using the concept of injectivity and surjectivity. (So start by constructing a function for the situation.)

1. Someone steps on your foot in the train.
2. Some people have multiple homes but others are experiencing homelessness.

[†] The symbol $!$ after a natural number means factorial, but those rarely arise in category theory.

[‡] In some specific situations injectivity might force surjectivity, or the other way round, say if the source and target set are both finite with the same number of elements.

3. In New York City, court records appear to be kept according to name and date of birth pairs. Lisa S Davis wrote about repeatedly being summoned to court for “unpaid tickets” that were not hers, and she eventually figured out that they were for another Lisa S Davis with the same birthday.[†]

Example 1 If someone steps on your foot then their foot has landed on yours. That’s two feet in the same place, which feels like injectivity has been violated. We can realize this by a function from the set of feet on the train to the set of positions on the train, with the function giving the position of each foot.

Example 2 It might be tempting to realize this one as people being mapped to homes, but that would require some inputs to have multiple outputs in order to capture the fact that some people have multiple homes; also people with no home would then have no output. Instead we can take the inputs to be homes and the outputs to be people. We need to be a bit subtle (or vague) about the fact that multiple people can live in a home, but glossing over that fact we now get that the failure of injectivity comes from some people having multiple homes, and the failure of surjectivity is from some people experiencing homelessness. (We might worry about abandoned houses that have no people living in them, but I would just not call it a “home” if nobody lives there.)

Example 3 Here we can take the set of inputs to be the people in New York City, and the outputs are pairs of information $\{name, date\}$. The function is not surjective: there are definitely dates on which nobody currently in NYC was born (for example, January 1, 1500). The story of Lisa S Davis ($\times 2$) is about the fact that this function is not injective, but unfortunately the DMV and NYPD[‡] seem to act on the assumption that the function *is* injective, and take this combination of information to be an identifier of a unique person.

The part of the story with a moral is that the tickets the author received were all for trivial infractions, and the other Lisa S Davis was Black. The Caucasian Lisa had never had a ticket for anything, but not because she’d never done anything. When she went to court to try and sort it out she was surrounded by black people who had been summoned for trivial offences. The judge didn’t believe that she wasn’t the correct Lisa S Davis, and she realized “For me, this was an inconvenience and an aberration. But I was beginning to understand that, for most of the people there, injustice was a given.”

[†] Lisa Selin Davis, “For 18 years, I thought she was stealing my identity. Until I found her.” *The Guardian*, April 3, 2017.

<https://www.theguardian.com/us-news/2017/apr/03/identity-theft-racial-justice>

[‡] Department of Motor Vehicles and New York Police Department.

Things To Think About

T 15.3 Now try these mathematical examples. See if you can get the intuitive idea, and then see if you can prove it rigorously if you like. But I think getting the intuitive idea is the important part. Again each one is a function $f: A \rightarrow B$.

- | | |
|---|--|
| 4. $A = \mathbb{Z}, B = \mathbb{Z}$
$f(a) = a + 1$ | 6. $A = \mathbb{Z}, B = \text{the even numbers}$
$f(a) = 2a$ |
| 5. $A = \mathbb{Z}, B = \mathbb{Z}$
$f(a) = 2a$ | 7. $A = \emptyset, B = \mathbb{Z}$
f is the empty function. |

Example 4 This function is bijective. Adding 1 to numbers is always injective, and now that we have no smallest number, every number n can be achieved starting from $n - 1$, so the function is also surjective.

Example 5 This function is injective: if $2a = 2a'$ then certainly $a = a'$. However it is not surjective; for example there is no integer a such that $2a = 1$.

Example 6 This function is injective by the same proof as the previous example, but it is now surjective as well, so it is a bijection. For the surjective part observe that every even number is of the form $2k$ for some $k \in \mathbb{Z}$, and $f(k) = 2k$. This gives us a bijection between all the integers and the even numbers. This is a little counterintuitive as you might think that the even numbers are just “half” of the integers. But this is one of the weird and wonderful things about infinite sets — you can take half of the set and still have the “same number of things”. We try not to call it a “number” at this point, because our intuition about numbers is very different from how infinities behave. This is really telling us that the even numbers and the integers give the same level or “size” of infinity. By contrast, it is a profound fact that there is no bijection between the integers and the real numbers, and this leads to the idea that there is a hierarchy of bigger and bigger infinities.

Example 7 The empty function is the one that is vacuous, and it might be hard to use intuition to decide if it is injective or surjective. Here the formal definition is what we need. For injective, the definition starts “for all $x, y \in A$ ”. We can now stop because A is empty, so whatever happens this condition is vacuously satisfied. Thus the function is injective. For surjective there is a part of the definition that says “ $\exists a \in A$ such that...” and then we’re in trouble because there is no element in A . The only way we could survive this is if the condition is vacuously satisfied, that is: the condition starts with “ $\forall b \in B$ ” so if B is empty then the condition is vacuously satisfied. Here B is not empty so it is not surjective, but the empty function $\emptyset \rightarrow \emptyset$ is both injective and surjective. As you might expect it is a bijection between the empty set and itself.

As we have seen, functions ostensibly given by the same formula can take on different characteristics depending on what we take their source and target sets to be. This is one of the reasons I prefer the categorical approach where functions always come with a source and target as part of their information. We are now going to give a categorical approach to injectivity and surjectivity.

15.3 Monics: categorical injectivity

The definitions of injectivity and surjectivity had elements of sets all over them. To express them categorically we need to express them using only sets and *functions*, not sets and *elements*. So we need to avoid saying anything like “for any element in A ” or “there exists an element in A ”.

We consider a morphism $a \xrightarrow{f} b$ in an arbitrary category $\mathcal{C}$.

Note on notation

I prefer lower case letters for objects of a category. I might use upper case if I know they’re sets and therefore have elements inside them which I may then denote with lower case letters. I know this sort of thing can be confusing when you’re learning new math, but it is essentially impossible to keep consistent notation across all of math. We can try and stick to some general principles, while also remaining flexible about what individual letters can denote.

Instead of considering pairs of elements being mapped to the same place, we can look at pairs of morphisms going in, as shown in this key diagram called a “fork”.

$$m \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \xrightarrow{f} b$$

Definition 15.4 A morphism $a \xrightarrow{f} b$ in a category $\mathcal{C}$ is called a *monomorphism* or *monic* if, given any fork diagram as above in $\mathcal{C}$,

$$f \circ s = f \circ t \implies s = t.$$

Note that the $\circ$ notation for composition gets a bit tedious and redundant so we might just write this condition as: $fs = ft \implies s = t$.

Now, this definition is supposed to be

1. a categorical version of injectivity. and
2. part of being an isomorphism.

So it would be sensible to check both of those things.

Things To Think About

T 15.4 Can you work out what it means to check both of those things, and then check them?

Checking that monics are a categorical version of injectivity amounts to checking that in the case of **Set** monics correspond to injective functions.

Proposition 15.5 *A function is injective if and only if it is monic in **Set**.*

Proof Consider an injective function $f: a \rightarrow b$. We aim to show f is monic.

So we consider functions s and t as in this fork diagram such that $fs = ft$, and want to show $s = t$.

$$m \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \xrightarrow{f} b$$

Note that I am going to be a bit lazy about brackets here. I will write $fs(x)$ to mean both $(fs)(x)$ and $f(s(x))$. The point now is that $s(x)$ and $t(x)$ are two elements mapped to the same place by the injective function f , so they must be the same.

Now $fs = ft$ means: $\forall x \in a \quad fs(x) = ft(x)$
 but f is injective, so this implies: $\forall x \in a \quad s(x) = t(x)$
 thus $s = t$

so f is monic as required.

Conversely suppose $f: a \rightarrow b$ is monic in **Set**. We aim to show that f is injective. So consider $x, y \in a$ such that $f(x) = f(y)$; we want to show $x = y$.

We need to re-express this situation using functions s and t into a as in the fork diagram so that we can invoke the definition of monic. We can do this by making m a one-element set: functions out of that will then just pick out one element of a .

Let 1 denote a set with one element $*$. We set up a fork diagram with s, t defined by $s(*) = x$ and $t(*) = y$.

$$1 \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \xrightarrow{f} b$$

We are now going to show that $fs = ft$, then invoke the definition of monic to conclude that $s = t$ which amounts to $x = y$ as we need. So we want to show that fs and ft do the same thing to every input — but the only input is $*$.

Now $fs(*) = f(x)$ by definition of s
 $= f(y)$ by hypothesis (i.e. using the thing we supposed)
 $= ft(*)$ by definition of t
 thus $fs = ft$ as functions (because they agree on every input).

But f is monic, so $s = t$, and so $s(*) = t(*)$. That is, $x = y$ as required. □

We'll now check that being monic is part of being an isomorphism.

Proposition 15.6 *Let f be a morphism in any category. If f is an isomorphism then f is monic.*

Proof Let $f: a \rightarrow b$ be an isomorphism with inverse g .

Consider the usual fork diagram. We aim to show that if $fs = ft$ then $s = t$. We have the following situation.

$$m \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{1_a} \end{array} b \xrightarrow{g} a$$

A category theorist might be satisfied at this point, by reading the diagram “dynamically”. Here are the steps of that reading.

$$\begin{aligned} fs = ft &\implies gfs = gft \\ &\implies s = t \quad \text{since } gf = 1_a \end{aligned}$$

Thus f is monic as required. $\square$

Reading a diagram dynamically means taking one part of the diagram at a time and moving through the diagram via some equalities. We will gradually build an understanding of how to do this.

Note we only used part of the definition of isomorphism, which is coherent with the fact that “monic” is a generalization of “injective”, which is only part of the definition of bijection. This part is interesting enough that it gets a name.

Definition 15.7 Let $f: a \rightarrow b$ be a morphism in a category.

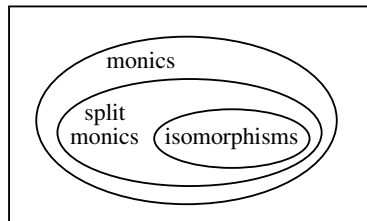
If there is a morphism g as shown here with $gf = 1_a$ then f is certainly monic (as in the proof above) and is called a *split monic*. Then g is called a *splitting*, *left inverse* or a *retraction* for f .

$$a \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{1_a} \end{array} b \xrightarrow{g} a$$

I confess I can never remember the convention on lefts and rights (it goes by algebraic notation rather than by arrows, which confuses me) so I find it safest to draw the diagram.

It is a slightly subtle point that not all monics are split. So we have a sort of Venn* diagram like this. Split monics are a particularly handy type of monic because they are stable under transfer to other categories. We’ll look at this idea of stability in Chapter 20.

morphisms in a given category



15.4 Epics: categorical surjectivity

Surjectivity is like injectivity but the other way round. The formal set-based (non-categorical) definitions don't look so much like "other way round" versions of each other, but with the categorical version we can see a very precise sense in which they are the other way round from each other: they are *duals*, which means that in the categorical definitions all we need to do is turn all the morphisms to point the other way. If we do this to the definition of monic we get the definition of epic, which is a categorical version of surjectivity.

Things To Think About

T 15.5 Write out the definition of monic again but with every arrow pointing the other way. This is the definition of "epic". Now see if you can prove that in **Set** a morphism is epic if and only if it is surjective.

If I very literally turn all the arrows around in the definition of monic the diagram will become this.

$$m \xleftarrow{s} a \xleftarrow{f} b$$

However, I prefer to keep the abstract shape but adjust the notation so that the morphism f still goes $a \rightarrow b$ and all arrows point to the right, as shown here.

$$a \xrightarrow{f} b \xrightleftharpoons[t]{s} e$$

Here is the resulting definition; I will fully acknowledge that it's hard to remember which direction is which out of monic and epic unless/until you work with them a lot.

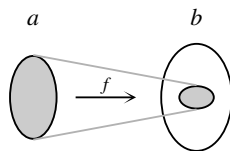
Definition 15.8 A morphism $f: a \rightarrow b$ in a category is called an *epimorphism* or *epic* if, given any diagram as above: $sf = tf \implies s = t$.

We're now going to show that the epics in **Set** are precisely the surjective functions. The proof proceeds analogously to the result for monics, the idea being to "translate" the properties of surjectivity into morphisms.

Proposition 15.9 A function f is surjective if and only if it is epic in **Set**.

This proof is going to be helped along by a few preliminary ideas.

The idea is that when you have a function $a \rightarrow b$ the target set b splits up into the part that is "hit" by f and the part that isn't. Here's an intuitive picture. The function f maps all of a into b somewhere, possibly not landing on all of it.



The part that the function f does land on is called the *image* of f .

Definition 15.10 The *image* of a function $f: a \rightarrow b$ is a subset of b defined as follows: $\text{Im}f = \{ f(x) \mid x \in a \}$.

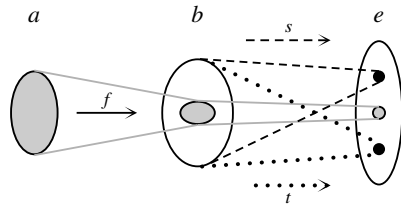
Here the vertical bar $\mid$ indicates “satisfying”, in the context of defining a set: the set consists of the things on the left of the bar, satisfying the conditions on the right. So this set is all elements $f(x)$; another, less succinct but perhaps more intuitive way of putting it, is: $\text{Im}f = \{ y \in b \mid \exists x \in a \text{ such that } f(x) = y \}$.

Now note that the set b then splits into two parts: the part that is hit by f and the part that isn't. I'll call the part that isn't hit the *non-image*, and write it NonIm . This is not standard, I just invented it for now. The set b is then the *disjoint union* of those two sets: $b = \text{Im}f \sqcup \text{NonIm}f$.

That is, everything in b is in the image or the non-image, and not both.

Now the situation we're looking at involves having two more functions that follow on from f , so they map the whole of b somewhere else. And the point is that if they are equal upon pre-composition with f then they must agree on the image of f . However, they could disagree on the non-image.

This picture is trying to capture that. The two further functions s and t both map the gray part of b to the same place, but map the outside to different places.



The essence of the proof is then that such an s and t can be different if and only if something exists in the non-image of f , otherwise there will be no space for them to be different while being the same when pre-composed with f . Incidentally I have implicitly used the following characterization of surjectivity.

A function $f: a \rightarrow b$ is surjective if and only if $\text{Im}f = b$.

We are finally ready to prove that a function is surjective iff it is epic.

Proof of Proposition 15.9 First we suppose that f is surjective and aim to show that it is epic.

So we consider functions s and t as in this fork diagram such that $sf = tf$, and want to show $s = t$.

$$a \xrightarrow{f} b \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} e$$

The idea is that f lands on all of b , so there's no extra room for s and t to be different given that they're the same when pre-composed with f .

Consider $y \in b$. Now f is surjective, so: $\exists x \in a$ such that $f(x) = y$.

$$\begin{aligned} \text{Thus } s(y) &= sf(x) \text{ by substitution} \\ &= tf(x) \text{ by hypothesis} \\ &= t(y) \text{ by substitution} \end{aligned}$$

so $s = t$ as functions, as required.[†]

For the converse we are going to prove the contrapositive, that is, that if f is not surjective then it is not epic.

The idea is that if f is not surjective then we can fabricate two functions s and t that agree on the image of f but disagree on the non-image. I'm going to take e to be a set of 3 objects so that there's a landing spot for where s and t are the same, and one landing spot each for where they're different. It doesn't matter what these three objects are so I just called them 1, 2, 3.

Define a set $e = \{1, 2, 3\}$, and define functions $s, t : b \rightarrow e$ as follows.

$$s(y) = \begin{cases} 1 & \text{if } y \in \text{Im}f \\ 2 & \text{if } y \in \text{NonIm}f \end{cases} \quad t(y) = \begin{cases} 1 & \text{if } y \in \text{Im}f \\ 3 & \text{if } y \in \text{NonIm}f \end{cases}$$

First note that since f is not surjective, $\text{NonIm}f$ is not empty, so $s \neq t$. However we are going to show $sf = tf$, that is: $\forall x \in a, sf(x) = tf(x)$.

Now, given any $x \in a$, we know $f(x) \in \text{Im}f$ (by definition), so $s(f(x)) = 1$. Similarly $t(f(x)) = 1$. So $sf = tf$, but $s \neq t$, so f is not epic. $\square$

This proposition gives rise to what is probably my favorite category theory joke:[‡] “Did you hear about the category theorist who couldn't prove that a function was surjective? Epic fail.”

In the previous section we showed that “isomorphism $\Rightarrow$ monic”, so we could now show the analogous result for epics.

Things To Think About

T 15.6 Look back at the proof of the result for monics and see if you can directly translate it into a proof for epics.

Proposition 15.11 *Let f be a morphism in any category. If f is an isomorphism then f is epic.*

[†] This wording might sound strange. We could just say “ $s = t$ as required”, but I put “as functions” in to remind ourselves that we have checked they are equal on each element, which means that the entire function s is equal to the entire function t .

[‡] I believe I first heard this joke from James Cranch.

more I'm not sure it's a good thing to get hung up on. I'd rather stress that the abstraction has various points:

1. It enables us to make connections with more situations other than just sets and functions, by now looking for these properties in other categories.
2. It gives us a sense in which injective and surjective are precisely “the other way round” from one another, by categorical duality.

We're now going to finish thinking about the relationship with isomorphisms.

15.5 Relationship with isomorphisms

We have seen that all isomorphisms are both monic and epic. However, the converse isn't true: if f is a *split* monic and a *split* epic then it has an inverse on both sides which is precisely the definition of isomorphism, but a morphism could be monic and epic without necessarily being an isomorphism, if it doesn't split. To exhibit this we will need to look at a category of objects with more structure than sets, because in **Set** we know that

1. monic = injective
2. epic = surjective
3. isomorphism = bijective = injective + surjective

So in **Set** it happens that “monic + epic $\Rightarrow$ isomorphism”. However, we can at least show that the splitting of epics is subtle, by considering infinite sets.

Things To Think About

T 15.7 We are going to think about a function that is surjective but might not have a splitting. We'll warm up with finite sets and then do infinite sets.

1. Let $f: \{-1, 0, 1\} \rightarrow \{0, 1\}$ be defined by $f(x) = |x|$. This function is surjective thus epic; can you find a splitting for it?
2. Let $f: \mathbb{Z} \rightarrow \mathbb{N}$ be defined by $f(x) = |x|$ as above. This function is surjective thus epic; can you find a splitting for it?

Remember that for f to be a split epic we need g as shown here.

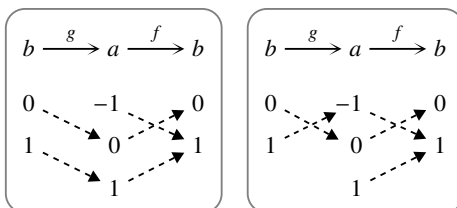
$$b \xrightarrow{g} a \xrightarrow{f} b$$

$\dashrightarrow$
 1_b

The idea is to try and make a function g going backwards, sending every element x to a *pre-image* of x under f , that is, an element that f maps to x . If f

maps several elements to the same place then we have a choice. So for the first example 0 has to go to 0 but 1 can go to 1 or -1 .

Here are pictures of those two possibilities. In either case if you follow the dotted arrows from the beginning to the end, 0 and 1 each go back to themselves, they just take a slightly different route in the middle.



Now for the second example it's the same idea but we are doing the entire integers. So 0 has to go to 0, but any other n can go to n or $-n$. This means that to construct a splitting $\mathbb{N} \rightarrow \mathbb{Z}$ we need to make that choice for each natural number. In this case it's not hard: we can just decide right from the start that we're going to send everything to the positive version, say.

However, if we were in a situation without a way to make one global choice then we would have to make an individual choice for each natural number. In case you've heard of it, I'll briefly mention that this is exactly what the Axiom of Choice is about: the question of whether or not it is possible to make an infinite number of choices, even if in each case the choice is only between a finite number of things. The Axiom of Choice says that it is possible, and this axiom is *logically equivalent* to saying that all epics split. There are different versions of set theory that include or don't include the Axiom of Choice. You needn't worry about it if you don't want to, it's just good to be aware if you are implicitly invoking it to make an infinite number of choices.

Finally here I'd like to stress a difference between property and structure. The definition of monic/epic is a *property* of a morphism, whereas a splitting is a piece of *structure*, and that's a technically and ideologically important difference. A property is a bit nebulous, whereas structure is something that we can specify and more likely carry around with us to other categories. It's a bit like saying someone is beautiful (a property) rather than being specific about a feature they have, like a warm smile, or a generous heart.

15.6 Monoids

In the category **Mnd** of monoids and monoid homomorphisms things are a little more subtle because of the requirement of preserving structure. This means we have less freedom over where to send elements, and some things end up being determined for us.

For example, consider the monoid of natural numbers $\mathbb{N}$ under addition. We will try to define a monoid homomorphism f to some other monoid M .

- We know that 0 is the identity in $\mathbb{N}$ so it must go to the identity in M ; we have no choice about that.
- We can send 1 anywhere we want.
- We know that $2 = 1 + 1$ in $\mathbb{N}$, and by the definition of monoid homomorphism we must have $f(1 + 1) = f(1) + f(1)$ so we have no choice about where 2 goes: it has to go to $f(1) + f(1)$.
- Something similar is now true for every other natural number. Every n is a sum of 1's and so we have no choice over where it goes: it has to go to the n -fold sum of $f(1)$'s.

What we've done here is based on the fact that $\mathbb{N}$ is the *free monoid* built up from a single element. We saw this in Section 11.2 when we started with one non-trivial arrow and built up a whole category from it, and we'll come back to it in Chapter 20 when we talk about free functors. For now I want to show that this forces more things to be epic than just the surjective homomorphisms.

Proposition 15.13 *Let $f: \mathbb{N} \rightarrow \mathbb{Z}$ be the monoid homomorphism given by the inclusion (that is, $f(n) = n$). Then f is epic in **Mnd**.*

Things To Think About

T 15.8 Have a think about monoid morphisms $\mathbb{Z} \rightarrow M$, in the same spirit as with $\mathbb{N}$ above. What choice do we have about where things go? If you're confused by M being a random monoid, try just taking it to be $\mathbb{Z}$.

Once you've worked that out, try and make monoid morphisms s, t as in the definition of epic, with $sf = tf$.

$$\mathbb{N} \xrightarrow{f} \mathbb{Z} \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} M$$

What choice do we have over s and t ?

Mapping out of $\mathbb{Z}$ forces our hand in much the same way that mapping out of $\mathbb{N}$ does. Deep down this is because $\mathbb{Z}$ only differs from $\mathbb{N}$ by having additive inverses, but where those go is forced by preserving the monoid structure, as we saw when we defined group homomorphisms. So we know that when we define s and t the only choice we have is about where 1 goes, and the only way they can be different is if they do different things to 1.

Now if we also have $sf = tf$ that means s and t agree on the image of f . However, $1 \in \mathbb{Z}$ is in the image of f (because $f(1) = 1$) and so we must have $s(1) = t(1)$, which in turn forces $s = t$. This shows that f is epic even though it isn't surjective.

Note that f is injective and this is enough to ensure it is monic in **Mnd**. So in fact f is monic and epic, but it is not an isomorphism.

15.7 Further topics

Density

There are many examples similar to monoids where morphisms are epic without being surjective, on account of the action of a morphism being determined by just part of it. I'll include some examples here just in case you've heard of them, to make some connections in your mind; if you haven't heard of them then it won't make much sense to you, but we won't use it anywhere.

For example, with topological spaces a continuous map is entirely determined by its action on a “dense” subset. A basic example is if we consider a closed interval such as $[0, 1]$ — remember this means it's a portion of the real line including the endpoints 0 and 1. If we define a continuous function from here to somewhere else, once we've defined it on the *open* interval $(0, 1)$ (that is, without saying what it does to 0 and 1) there is no choice left about what it does to 0 and 1. A set A is called dense inside B if the “closure” of A is B ; the closure of A is essentially the most economical way to enlarge A just enough to make it a closed set.

This is related to how we can construct the real numbers from the rationals, by taking the closure under limits, which amounts to throwing in the irrational numbers. In general B is the closure of A if the only thing B has extra is limits for sequences in A that really “want” to converge but didn't have a limit. Continuous functions preserve limits, so their action on B will be entirely determined by the action on A .

This sort of thing happens any time something has been constructed “freely” from some generators. If the morphisms in that category preserve the structure, then all you have to know is where the generators go, and then there's no choice of where the rest of the structure can go. You might also have seen this with vector spaces and basis elements. There's a concept of “limit” and “density” in category theory that mimics the situation of limits and density in analysis.

Subobjects

One special case of injective functions is subset inclusions. For example we have seen the function $\mathbb{N} \rightarrow \mathbb{Z}$ which just takes each natural number and sends it to the same number as an integer. A subset inclusion is when A is a subset[†]

[†] A is a subset of B if all the elements of A are elements of B . It could be the whole of B or it could be empty, or it could just have some of the elements of B .

of B and all you do is map each element of A to “itself” in B . Now this is an un-categorical notion because we’re talking about *elements* of sets, and moreover we’re asking for elements of A to be *equal* to some elements of B .

There’s no categorical reason that a subset inclusion is any better than any other kind of monic, and so in more general situations monics are used to indicate *subobjects*. This helps us account for the possibility that something might have different labels but apart from that be really like a sub-thing. This makes less sense with sets because they have no structure; it might seem odd to think of the set $\{1, 2\}$ as a subset of the set $\{a, b, c\}$ but it’s a little less odd to think of a cyclic group of order 2 as a subgroup of a cyclic group of order 4 even if the elements have different names.

Higher dimensions

Monic and epic are categorical versions of injective and surjective functions, and when we go into higher dimensions we want to generalize injective and surjective functions to the level of *functors*, that is, the morphisms in the category **Cat**. We can try looking for monics and epics in that category, but it won’t really give us “the right thing” because that is working at the wrong dimension. **Cat** is really trying to be a 2-category (as we mentioned before) with a more subtle type of relationship between morphisms. That means that we shouldn’t be asking for morphisms to be equal, as we do in the definition of monic/epic. In Chapter 21 we will see that we want something more like “injective and surjective up to isomorphism” and that this gives the concepts of full and faithful functors.